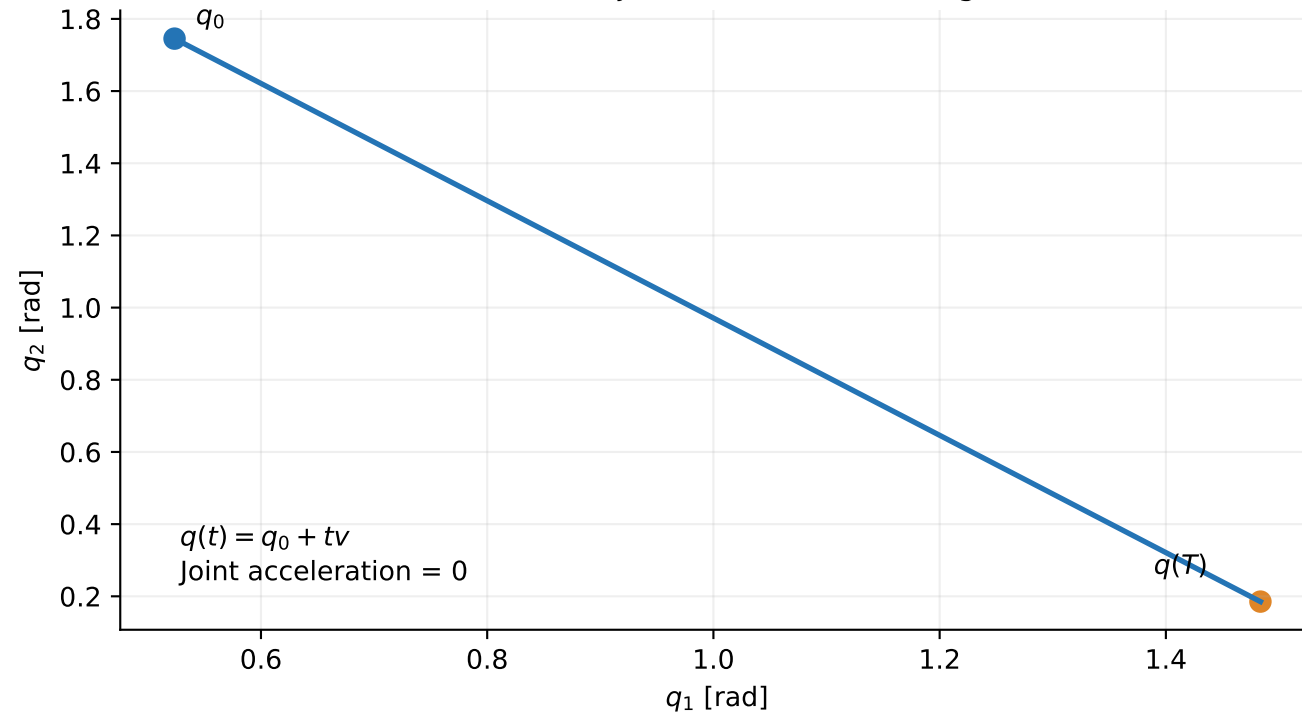
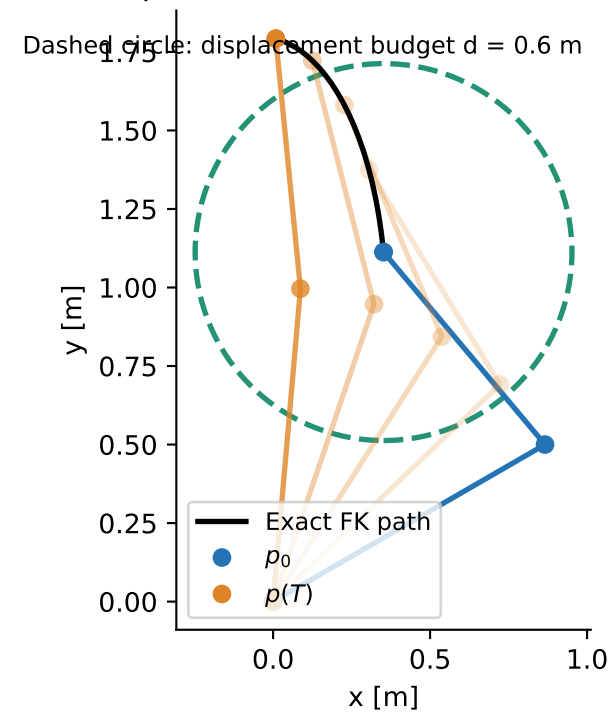


Two-link planar arm: our kinematic bounds versus exact FK

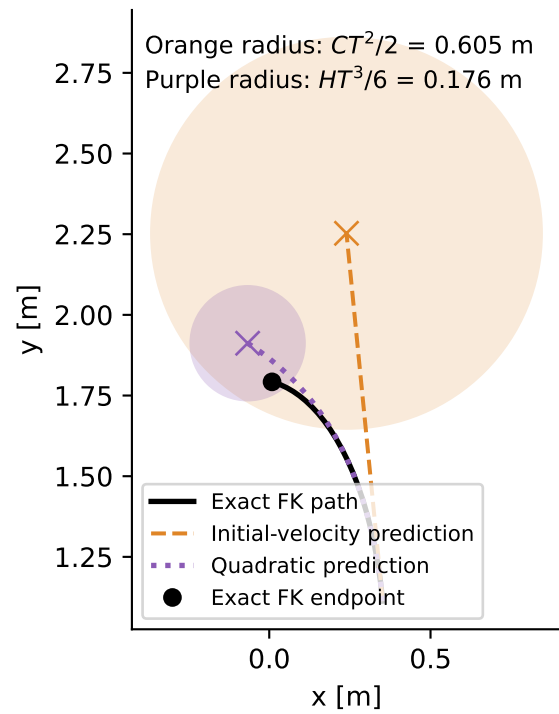
A The nominal joint motion is a straight line



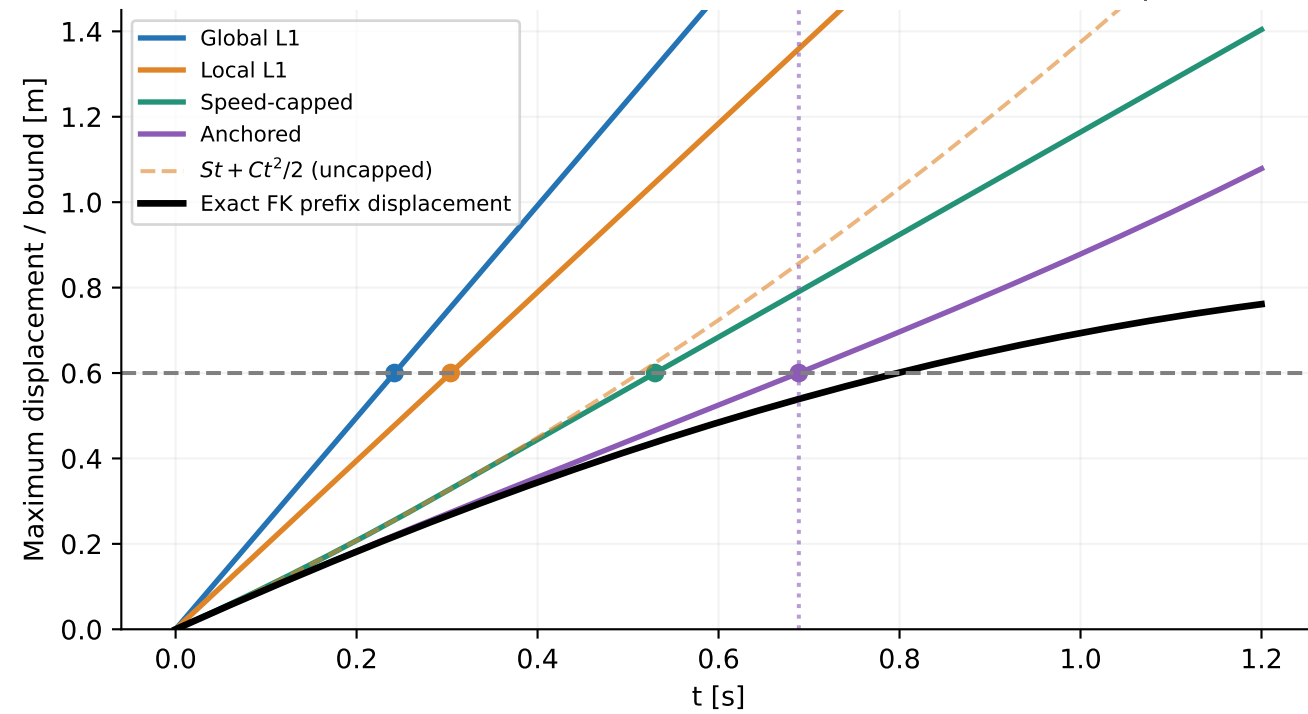
B FK maps that line to a curved workspace path



C What the bounds know about the endpoint at T



D Same motion, same clearance: when does each stop?



Lengths = (1.0, 0.8) m • $q_0 = (30^\circ, 100^\circ)$ • $v = (0.8, -1.3)$ rad/s • $T = 1.2$ s
End-effector displacement only; the clearance circle is a hypothetical world-safety budget, not whole-arm collision checking.